

Bayes theorem

→ Nav Bias Algo in ML

- Bayes theorem helps you update your belief when new information arrives.

→ "give some new evidence, how should i change my thinking?"

→ Formula :-
$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

where :- $P(A|B)$ $\Rightarrow$ Probability of A given B
[This is what we want]

$P(B|A)$ $\Rightarrow$ probability of B given A (likelihood)

$P(A)$ $\Rightarrow$ probability of A (prior belief)

$P(B)$ $\Rightarrow$ probability of B (overall chance of B)

→ For ex:-

updated Belief = (How likely the evidence is if A is true
X
old belief about A)

How common the evidence is.

Examples :-

① Rain & cloud

• probability it rains $= P(\text{Rain}) = 0.2$ (20%)

• probability of clouds if it rains $= P(\text{cloud} | \text{Rain})$

$$P(\text{clouds} | \text{Rain}) = 0.9 \text{ (90\%)}$$

• probability of clouds in general

$$= P(\text{clouds}) = 50\%$$

Q) If you see clouds, what is the probability it will rain?

$$\Rightarrow P(\text{Rain} | \text{cloud}) = \frac{0.9 \times 0.2}{0.5} = \frac{0.18}{0.5} = 0.36$$

$\Rightarrow$ 36% chance of rain when you see clouds.

$\rightarrow$ Before cloud $\Rightarrow$ 20% chance.

$\rightarrow$ After cloud $\Rightarrow$ 36% chance.

② Medical test :-

1% people have a disease $\rightarrow P(\text{Disease}) = 0.01$

Test is 99% accurate $\rightarrow P(\text{positive} | \text{Disease}) = 0.99$

False positive rate = 5% $\rightarrow P(\text{positive} | \text{No disease}) = 0.05$

Q) If a person tests positive, what is the probability they actually have the disease?

$\rightarrow$ Step - 1 :- Find $P(\text{positive})$.

$$\begin{aligned} \Rightarrow P(\text{positive}) &= (0.99 \times 0.01) + (0.05 \times 0.99) \\ &= 0.0099 + 0.0495 = 0.0594 \end{aligned}$$

$\rightarrow$ Step 2 :- Apply Bayes.

$$P(\text{Disease} | \text{positive}) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} \approx 0.166$$

$\rightarrow$ only 16% chance of actually having disease.

$\rightarrow$ Bayes is extremely important in ML, medical diagnosis, fraud detection, spam filtering etc.

→ What is the use of Bayes in DS?

① Spam Detection :-

A = Email is spam.

B = Contains word "Free"

→ $P(\text{Spam} | \text{Free})$

② Recommendation System :-

A = User like action movies

B = user watch 3 action trailers.

→ Bayes updates belief that user prefers action.

③ Naive Bayes Classifier :-

→ Very popular ML algorithm based fully on Bayes theorem.

→ Text classification

→ Sentiment analysis.

→ Email filtering.